



E-Commerce Customer Churn Prediction

End-to-end machine learning project predicting customer churn for an e-commerce platform.

Tools: Python · Scikit-learn Pipeline · GridSearchCV · SHAP · KNN Imputer · GitHub



Project Overview

This project simulates a real analytics mission: identifying which customers are likely to stop purchasing — and why — so the business can act before it's too late.

Business question:

"Which customers are at risk of churning, and what actions can we take to retain them?"

Why churn matters:

Acquiring a new customer costs 5x more than retaining an existing one. Predicting churn early gives the business a window to intervene with targeted retention strategies.



Dataset

Property	Value
Source	E-Commerce Customer Churn Dataset (Kaggle)
Rows	5,630 customers
Features	20 columns
Target	Churn (0 = Retained · 1 = Churned)
Class balance	83% Retained · 17% Churned



Key Insights

-  **Tenure is the #1 predictor** — new customers (low tenure) churn significantly more than long-term customers. The median tenure for churners is ~3 months vs ~10 months for retained customers
-  **Complaints drive churn** — customers who filed a complaint in the last month are far more likely to leave, making complaint resolution a critical retention lever

- 📦 **Recency matters** — customers who haven't ordered recently (high DaySinceLastOrder) show strong churn signals, confirming the classic RFM behavior
 - 💰 **Cashback protects loyalty** — customers receiving lower cashback amounts churn more, suggesting that reward programs have a measurable retention effect
 - 👤 **Single customers churn more** — MaritalStatus_Single showed a meaningful churn signal, possibly reflecting less stable purchasing habits
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🔧 ML Workflow

1. Data Quality

- **7 columns** had missing values (4.5-5.5% each)
- Strategy: **KNN Imputer (k=5)** — chosen over mean/median to avoid distortion from outliers, and over row deletion to preserve information
- Outliers kept — values like 127km warehouse distance or 61 months tenure are realistic edge cases

2. Feature Engineering

- **One-Hot Encoding** applied to 5 categorical variables (PreferredLoginDevice, PreferredPaymentMode, Gender, PreferredOrderCat, MaritalStatus)
- Result: 20 → 36 features after encoding

3. Modeling — Sklearn Pipeline

```
python

pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('model', RandomForestClassifier(class_weight='balanced', random_state=42))
])
```

Why Pipeline? Prevents data leakage by applying transformations consistently across train/test splits and future production data.

Why `class_weight='balanced'`? The 83/17% imbalance is moderate — no need for synthetic data (SMOTE). The balanced weight adjusts the loss function internally without distorting the data distribution.

4. Evaluation

Metric	Class 0 (Retained)	Class 1 (Churned)
Precision	0.96	0.97
Recall	0.99	0.78
F1-Score	0.98	0.87
Accuracy	0.96	

5. Cross-Validation (5-fold)

Fold	F1 Score
Fold 1	0.954
Fold 2	0.891
Fold 3	0.930
Fold 4	0.930
Fold 5	0.885
Mean	0.918
Std	0.026

Low standard deviation (0.026) confirms model stability — results are not due to a lucky split.

6. Hyperparameter Optimization — GridSearchCV

```
python
param_grid = {
    'model__n_estimators': [100, 200, 300],
    'model__max_depth': [5, 10, None]
}
```

45 fits (9 combinations × 5 folds). Best params: `n_estimators=100, max_depth=None` — confirming default configuration was already optimal.





SHAP Explainability

SHAP values reveal not just *what* matters, but *how* each feature impacts individual predictions.

Top features by SHAP importance:

Rank	Feature	Direction
1	Tenure	Low tenure → churn ↑
2	Complain	Complaint filed → churn ↑
3	DaySinceLastOrder	Many days inactive → churn ↑
4	CashbackAmount	Low cashback → churn ↑
5	MaritalStatus_Single	Single → churn ↑



Tools Used

Category	Tools
Language	Python 3
Data Analysis	Pandas, NumPy
Visualization	Matplotlib, Seaborn
ML & Preprocessing	Scikit-learn (Pipeline, GridSearchCV, KNNImputer, StandardScaler, RandomForest)
Explainability	SHAP
AI Assistance	Claude AI, GitHub Copilot
Version Control	Git, GitHub



How to Run

```
bash
```

```
# 1. Clone the repository
git clone https://github.com/FernandDevCode/ecommerce-churn.git
cd ecommerce-churn

# 2. Install dependencies
pip install pandas numpy matplotlib seaborn scikit-learn shap

# 3. Run the notebook
jupyter notebook notebooks/01_exploration.ipynb
```

Project Structure

```
ecommerce-churn/
|
├── data/
|   └── ecommerce_churn.xlsx      # Raw dataset
|
├── notebooks/
|   └── 01_exploration.ipynb     # Full EDA + Pipeline + SHAP
|
├── docs/
|   └── business_recommendations.docx # Business report
|
└── README.md
```

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Built as part of a data analytics portfolio targeting the Canadian job market.